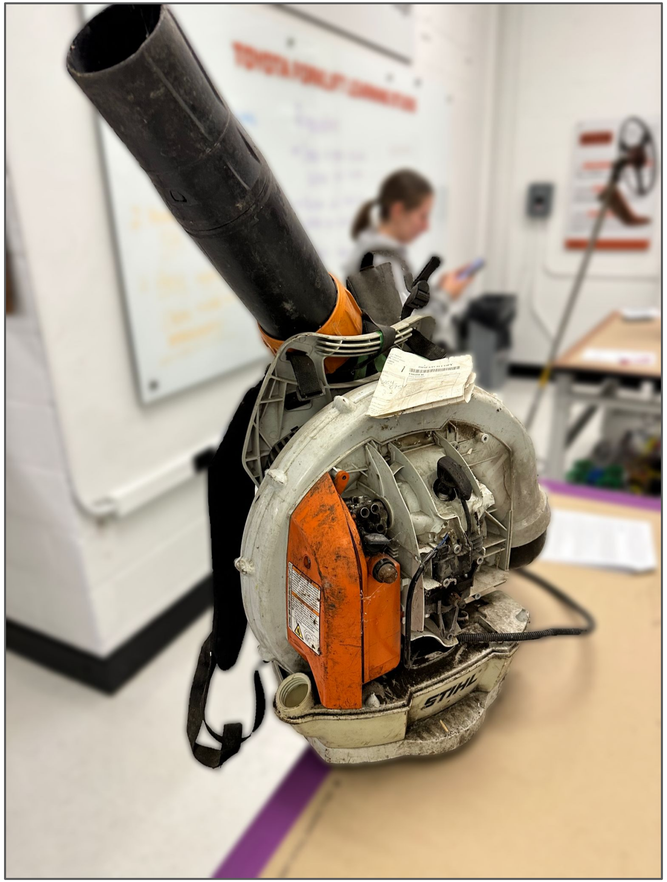
# Dissection of a Stihl BR 700 Leaf Blower





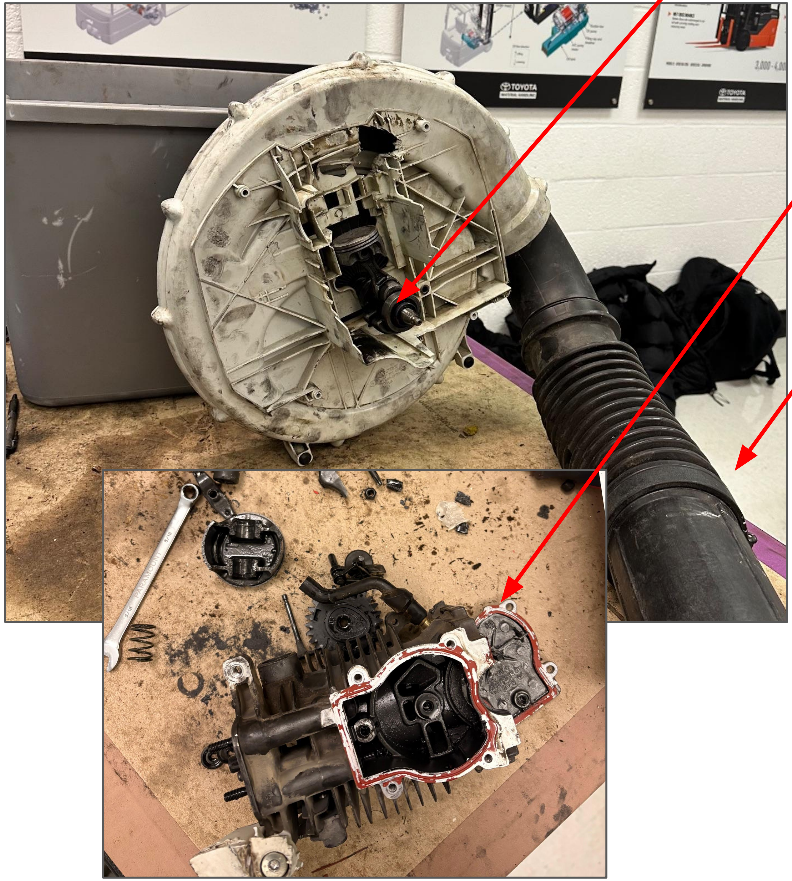
Fred Wentzien, Justice Rose, N
Li, Justin Tsemekhin

# Overview

Crankshaft (engine interior)

nozzle

engine block



impeller

volute

air intake side

Contributed by Fred Wentzien

# Key Airflow Components

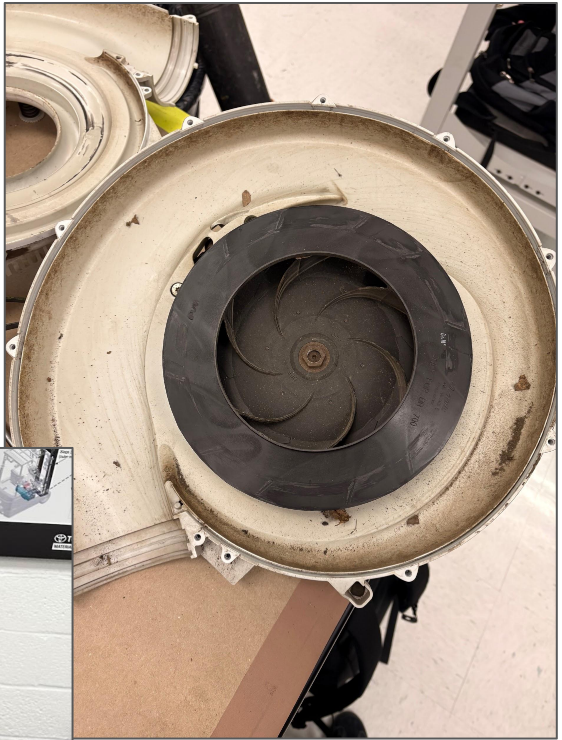
- Air intake at back
  - Determined by engine rpm
  - Air drawn into the center of radial flow pump
- Radial Flow Pump
  - Archimedean Spiral
  - The impeller accelerates air radially outward
  - High-pressure air at outer edge of spiral
- Converging Nozzle
  - Airflow enters nozzle
  - Pressure is converted to velocity
  - Specifications
    - Exit Velocity: 74m/s
    - Air flow rate: 1550 $m^3/h$

Conservation of Mass, Conservation of Momentum, Euler's Equation: The rotation of the impellers created a radial pressure gradient, which imparts kinetic energy into the fluid. This creates a low-pressure area in the center, drawing in air from the intake.

Bernoulli Principle: The constant streamlines in the nozzle allow us to determine the airspeed generated at the exit. As air flows through the nozzle, pressure decreases, causing an increase in velocity that can be calculated using Bernoulli's equation for streamlines.  Due to a low Mach number in the nozzle, we are able to assume incompressible flow, which allows us to use Bernoulli's equation.

# Radial Flow Pump





# Pump Characteristics

Bernoulli's Equation

$$p_{pump} + \frac{1}{2}\rho u_{pump}^2 + \rho g z = p_\infty + \frac{1}{2}\rho u_{out}^2 + \rho g z$$

$$\Delta p_0 = p_{pump} - p_\infty = \frac{1}{2}\rho u_{out}^2 = \frac{1}{2}(1.184)(74)^2 = 3241.792 \text{ Pa}$$

$$\boxed{p_0 = 3241.792 \text{ Pa}}$$

Specific speed (normal operation 74 m/s)

$$w^* = \frac{w Q^{1/2} \rho^{3/4}}{\Delta p_0^{3/4}}$$

$\omega \approx 4000 \text{ rpm} \cdot \frac{2\pi}{60} = 418.88 \text{ rad/s}$

$Q = 1550 \text{ m}^3/\text{hr} = .4306 \text{ m}^3/\text{s}$

$\rho = 1.184 \text{ kg/m}^3$

$\Delta p_0 = 3241.792 \text{ Pa}$

$$w^* = \frac{(418.88)(.4306)^{1/2}(1.184)^{3/4}}{(3241.792)^{3/4}}$$

$$\boxed{w^* = 0.726 \text{ at normal operating conditions}}$$

Specific speed (max flow 88 m/s)

$$w^* = \frac{w Q^{1/2} \rho^{3/4}}{\Delta p_0^{3/4}}$$

$\omega \approx 7200 \text{ rpm} \cdot \frac{2\pi}{60} = 753.98 \text{ rad/s}$

$Q = 1860 \text{ m}^3/\text{hr} = .5167 \text{ m}^3/\text{s}$

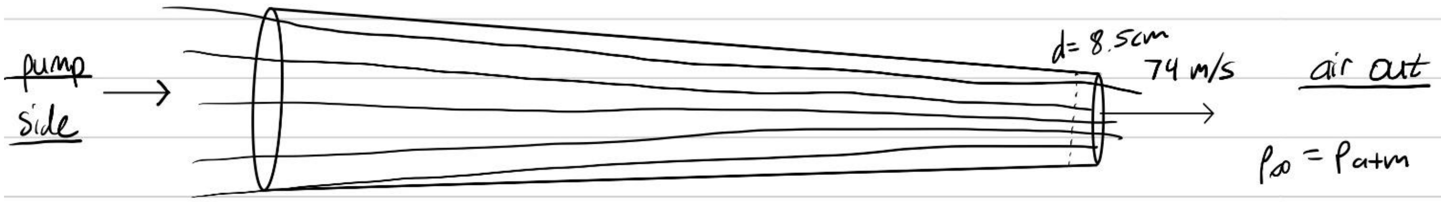
$\rho = 1.184 \text{ kg/m}^3$

$\Delta p_0 = \frac{1}{2}\rho u_{out}^2 = \frac{1}{2}(1.184)(88)^2 = 4584.448 \text{ Pa}$

$$w^* = \frac{(753.98)(.5167)^{1/2}(1.184)^{3/4}}{(4584.448)^{3/4}}$$

$$\boxed{w^* = 1.104 \text{ at extreme conditions}}$$

# Converging Nozzle Effect




Conservation of mass

$$\rho_{air} U_{pump} A_{pump} = \rho_{air} U_{out} A_{out}$$

$$U_{pump} = \frac{A_{out}}{A_{pump}} U_{out} = \frac{\left(\frac{8.5}{200}\right)^2 \pi}{\left(\frac{10.5}{200}\right)^2 \pi} 74 m/s$$

$$\boxed{U_{pump} = 48.5 \ m/s}$$

# Ideal Performance without loss

Avg $Re = 3.642e5$

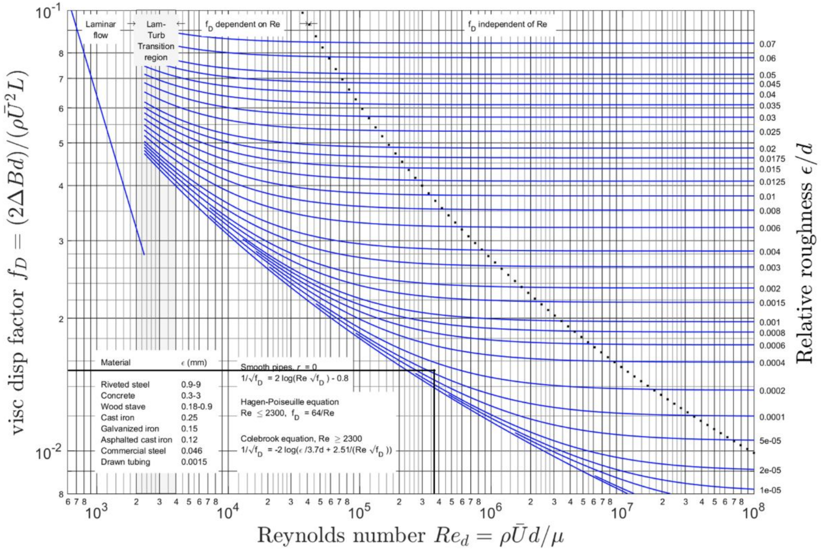



Chart:

$\frac{\epsilon}{d} = \frac{0.0015e-3}{0.095} = 1.6e-4$

$f_D = 1.65e-2$

$$\frac{\partial B}{\partial s} = \frac{1}{2}\rho U^2 f_D \frac{L}{d}$$, estimate $L = 3.5\text{ ft} = 1.067\text{ m}$

avg diameter $= 9.5\text{ cm} = 0.095\text{ m}$

avg $U = 61.25\text{ m/s}$

$$= \frac{1}{2}(1.2)(61.25)^2(1.65e-2)\frac{(1.067)}{(0.095)}$$

$$= 417.15\ Pa = \Delta p_{losses}$$

$$3241.79 = \Delta p_{pump}$$

ideal velocity

total $\Delta p = 3658.94\ Pa$

Bernoulli

$$p_1 + \cancel{\tfrac{1}{2}\rho U_0^2}^{\to 0} + \cancel{\rho g z}^{\to 0} = p_2 + \tfrac{1}{2}\rho U_{ext}^2 + \cancel{\rho g z}^{\to 0}$$

$$\Delta p = \frac{1}{2}\rho U_{exit}^2$$

calculate $U_{exit} = U_{ideal}$

$$U_{ideal} = \sqrt{\frac{2\Delta p}{\rho}}$$

$$= \sqrt{\frac{2(3658.94)}{1.2}}$$

$$\boxed{U_{ideal} = 78.10\text{ m/s}}$$

Manufacturer's velocity: 74.0 m/s
Ideal, lossless velocity: 78.1 m/s

*fín*